

Memora — Persistent Context Engine for Autonomous SRE

Anvil P-02 · L3 Final Writeup · release tag *anvil-2026-p02-L3-final*

L3 Weighted Automated	Latency p95 (worst seed)	Remediation Accuracy	Release Tag
0.3039 / 0.80	63 ms	44.8 %	anvil-2026-p02-L3-final
38.0% of max	vs 2000 ms budget	across 5 seeds	run: 2026-05-16

Axis	Weight	Score	Contribution
recall@5	0.30	0.1440	0.0432
precision@5_mean	0.15	0.1408	0.0211
remediation_acc	0.20	0.4480	0.0896
latency_p95 (63 ms vs 2000 ms)	0.15	1.0000	0.1500
manual_context	0.10	panel	—
manual_explain	0.10	panel	—
Weighted automated	0.80		0.3039

Per-seed breakdown · 5 council seeds × 25 eval signals = 125 reconstructions

Seed	recall@5	p@5	rem_acc	p95 ms	mean ms
314159	0.120	0.120	0.440	47	46.2
271828	0.080	0.080	0.400	63	48.1
161803	0.200	0.200	0.640	47	45.0
141421	0.160	0.152	0.360	62	45.0
173205	0.160	0.152	0.400	47	45.0

1. Memory Representation

Memora treats operational telemetry as evidence for long-lived incident memory, not as isolated records for keyword search. Every event is normalized at ingest into a stable envelope of timestamp, kind, service, canonical service id, incident id, trace id, extracted entities, attributes, and provenance hash, then routed into three structures formed during ingestion:

- a **service alias graph** that records every rename as an undirected edge and exposes BFS-resolved canonical lookup,
- a **per-incident profile** (`_IncidentProfile`) that accumulates nearby deploys, anomaly metrics, failure logs, traces, and remediations,
- a **service-keyed event index** for sub-millisecond temporal candidate generation at query time.

Profiles are formed during ingestion, not lazily during query — the L3 harness ingests the entire training corpus once and then asks reconstruction queries only for held-out signals. The profile produces a topology-independent **behavioural signature**:

shape_key = deploy | <5m | latency | timeout | callee | rollback | resolved

Two incidents with the same shape key are operationally equivalent regardless of service names. There is no embedding model, no ANN index, no learned retriever — every match is reproducible from the inputs and fully inspectable.

2. Relationship Synthesis

Reconstruction is a five-stage pipeline executed inside `Engine.reconstruct_context`:

- **Candidate generation** — pull events from a 30-min pre-signal / 15-min post-signal window via `_by_service` lookups against the alias-resolved canonical and the full alias closure.

- **Related-event ranking** — score by lineage match, alias overlap, incident-id continuation, entity mention, anomaly strength, and temporal proximity to the signal.
- **Causal chain synthesis** — emit ordered evidence edges: deploy → metric spike, deploy → failure log, metric spike → upstream failure, trace caller↔callee, failure log → remediation. Each edge stores cause id, effect id, evidence label, confidence, and an ordering proof.
- **Similar-incident matching** — for every historical profile, compute (a) shape-key score and Jaccard-weighted component score across metric_class, log_class, trace_class, remediation_types, delay_bucket, has_deploy, has_remediation, resolved; (b) lineage score from canonical / alias overlap; (c) trigger-token agreement.
- **Remediation transfer** — rank historical remediations by lineage match, shape match, age decay, and the success/failure ledger; remap the target through the alias graph so a payments - svc rollback is transferred to the renamed ledger - v2. The whole pipeline is bounded — fast mode caps related events at 10 and similar incidents at 5, which is what keeps the L3 worst-seed p95 at **63 ms**.

3. Drift Handling Strategy

The single biggest test in P-02 is whether the engine recognises a recurring family when the underlying service has been renamed. L3 makes this strictly harder by enabling **cascading renames** — the same canonical service can be renamed two to four times across the 21-day timeline:

```
payments-svc → billing-svc → ledger-v2 → ledger-v2-r5
              (all four resolve to one canonical via BFS)
```

Memora handles this with three layers:

- **Alias graph BFS.** _AliasGraph stores renames as undirected edges. resolve() and aliases() both run BFS from any input name, so every historical alias of a renamed service resolves to the same canonical and contributes evidence regardless of how many rename hops separate the train-time and eval-time names.
- **Behavioural shape matching.** When the alias chain breaks entirely (dependency drift renames the upstream caller, not the failing service), the topology-independent shape key still allows the historical incident to be found by behaviour alone.
- **Trigger-token inference.** When a signal arrives with only an alert string (alert:svc-04-r3/latency_p99_ms>3000), _infer_service_from_trigger extracts the service token before any service-keyed lookup, so even bare signals participate in alias resolution.

Contradictory remediation history is preserved rather than overwritten. Worked outcomes increase confidence; failed outcomes reduce it; older evidence decays. Each suggested remediation carries transfer proof: basis incident id, success/failure counts, target remapping through the alias graph, and the confidence contributors.

4. Latency Engineering

The benchmark adapter is intentionally stdlib-only, single-process, CPU-only — no embedding service, no graph database call-out, NumPy is used only by the harness. Three engineering decisions keep latency far under budget on the L3 stretch config (30 services × 21 days × ~76k events × 25 query signals per seed):

- **Indexes built at ingest.** _by_service, _by_incident, and _profiles are built incrementally during ingest(), so reconstruction never scans the full event log.
- **Bounded windows.** Fast mode uses a 30-min pre-signal / 15-min post-signal window. Even with cascading aliases, candidate generation touches a small slice of state.
- **Cached signatures.** _IncidentProfile.signature() memoises the shape key and component sets, so each historical incident is re-scored in microseconds.

Empirical L3 latency, across 5 seeds × 25 signals = 125 reconstructions:

```
worst-seed p95    =    63 ms      (vs 2000 ms fast-mode budget)
mean across all   =   45.9 ms
latency axis      =   1.000      (min(1, budget / worst-seed p95))
```

This earns the full 0.150 latency contribution and leaves the remaining axes as the optimisation surface.

5. Continuous Learning & Auditability

- **Profile formation.** Every new event recomputes the affected profile's signature; remediation events flip resolved and append to the historical ledger.
- **Feedback persistence.** _feedback stores every remediation observation with (incident_id, action, target, outcome, observed_at, canonical_service_id). Future similar incidents reweight remediation candidates against this ledger.

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- **Reasoning audit.** Every match in the returned Context carries lineage proof, shape-component scores, sequence audit, behavioural signature components, remediation history, and confidence contributors. The same Context flows through to the React workspace, so an operator inspects exactly what the engine did. This is the substrate the panel-graded `manual_context` and `manual_explain` axes evaluate.

6. Reproducibility

Single command, fixed seeds, locked stretch generator, deterministic adapter:

```
cd bench-p02-context
pip install -r requirements.txt
python run.py --adapter adapters.memora:Engine --out l3_report.json
```

Adapter source: `bench-p02-context/adapters/memora.py`. Rerunning on judges' machines reproduces the per-seed and aggregated numbers above to the rounding precision shown in `metrics.aggregate`.

7. Honest Assessment

Latency and remediation transfer behave well at the L3 stretch scale (latency at full credit; remediation at 44.8% across 5 action types — well above the 20% random baseline). Recall and precision are the active optimisation surface: the engine currently lands `recall@5 = 0.144` and `precision@5_mean = 0.141` against the L3 stretch (cascading renames + 20% decoy rate). The alias-graph BFS, shape signature, and trigger inference are all in place; the next iteration tightens candidate generation across deeper rename hops and the decoy-confidence threshold so unmatched signals return empty rather than confidently wrong. The architecture and the audit trail do not change — only the ranking calibration does.